KUET · ESE 4007 · Smart Grid Technology

ESE 4007 Question Map

Every question from the 2020, 2022, 2023 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 4007 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 4 h for Part A and 4 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×3, 2022 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: B4 (swing equation and equal area) is the one hard topic that keeps coming back; the power-angle graph makes it quick to understand. B5 (2020 numerical) appeared only once, in the oldest paper, so learn it last.

Section A: Smart grid concepts, distribution, communication, metering, PMU, load flow and analysis tools

Block 1: very likely and easy · 2.9 h

Topic	Chance	Level	Covered
A1 What a smart grid is: definition, conventional grid vs smart grid, architecture and components, key drivers, stakeholders, need for Bangladesh	Very likely 100%	Easy	12%
A3 Enabling technologies: information and communication; sensing, measurement, control and automation; power electronics and energy storage	Very likely 100%	Easy	24%

A6 Smart meter, AMR vs AMI, how AMI works, architecture for communication between smart meter and utility server	Very likely 100%	Easy	34%
A9 Load flow in smart grids: classical methods, old vs desired load flow, essential features, why needed and challenges	Very likely 100%	Easy	42%
A4 Distributed generation (DG) and microgrid: definition, advantages, configuration, grid-connected and islanded operation	Very likely 83%	Easy	53%
A7 Demand side management (DSM): load shifting, valley filling, peak clipping (load-shape curves)	Very likely 83%	Easy	60%
A2 Active vs passive distribution network; active distribution management scheme (block diagram)	Very likely 83%	Easy	67%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
A5 Smart grid communication: SCADA, HAN/NAN/WAN, and where Bluetooth, ZigBee, WLAN, WPAN and WiMAX are used	Very likely 100%	Medium	78%
✔ Pass line: 78% of Section A marks covered, enough to choose 3 of the 4 questions (about 3.7 h)			
A10 Performance analysis tools: contingency analysis, security-constrained OPF, DSOPF, static security assessment (SSA), congestion management; GIS	Very likely 100%	Medium	89%
A8 Phasor measurement unit (PMU) and synchronised phasor measuring system (WAMS)	Very likely 83%	Medium	95%
A11 Multi-agent systems (MAS): IDAPS and decentralised meter reading	Very likely 83%	Medium	100%

Section B: Power system control and stability, power quality, intelligence, security, renewables

Block 1: very likely and easy · 2.5 h

Topic	Chance	Level	Covered
B8 Computational intelligence and decision tools: ANN, expert systems, AHP, decision support tools, stability tools and computational challenges	Very likely 100%	Easy	18%
B9 Interoperability and cyber security: threats, false data injection attacks, mitigation	Very likely 100%	Easy	30%
B3 Reactive power and voltage control: why the grid needs reactive power, control methods, AVR, SVC, improving P_{max}	Very likely 100%	Easy	40%
B11 Barriers and solutions, standards and research areas, the simplest and the most complex thing about the smart grid	Very likely 100%	Easy	48%

B10 Renewables and the smart grid: integration challenges, education and training, V2G, sustainable energy options, efficiency and DG control	Very likely 83%	Easy	57%
B7 Power quality: issues in a smart grid, importance of power-quality management and audit	Very likely 83%	Easy	61%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
B2 Frequency control: why grid frequency matters, turbine speed governing system, load frequency control loop, AGC	Very likely 100%	Medium	72%
B6 Voltage stability: why instability occurs, preventive control, assessment techniques (P-V, Q-V, VIPI, indices)	Very likely 100%	Medium	83%
✔ Pass line: 83% of Section B marks covered, enough to choose 3 of the 4 questions (about 4.0 h)			
B1 Buses and bus types, infinite bus, bus admittance matrix Y_{bus} and proof of $I_{bus} = Y_{bus}V_{bus}$	Very likely 83%	Medium	88%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B4 Rotor angle stability: steady-state vs transient, swing equation, small-disturbance solution $\omega_n = \sqrt{(\pi f_0 P_s/H)}$, equal area criterion	Very likely 83%	Hard	98%

Only if time is left · 1.0 h

Topic	Chance	Level	Covered
B5 Numerical: voltage behind transient reactance and transient power-angle equation (with and without saliency)	Possible 17%	Hard	100%

Pass plan

Papers available: 2023 and 2022 (4 questions per section, answer 3; full marks 210) and 2020 (short paper: 3 questions of 30 marks per section, answer 2). No 2021 or 2024 paper was found in the question bank. Section A = smart grid basics, active distribution, enabling technologies, DG and microgrid, communication (SCADA, HAN/NAN/WAN), smart meter and AMI, DSM, PMU, load flow, performance analysis tools (contingency, SSA, SCOPF, DSOPF), multi-agent systems. Section B = buses and Y_{bus}, frequency control and AGC, reactive power and voltage control (AVR, SVC), rotor-angle stability (swing

equation, equal area criterion), voltage stability, power quality, computational intelligence (ANN, expert systems, AHP), cyber security, renewables and V2G, barriers and standards. There is no teacher material folder for this course, so the answers follow the standard textbooks the questions come from (J. Momoh, "Smart Grid: Fundamentals of Design and Analysis"; Saadat/Kundur for power system stability and control). All diagrams in this pack were drawn specially for it. Only one numerical exists (2020 4b, synchronous machine transient emf); the 2022 6c derivation of ω_n is the other mathematical question.

SECTION A: SMART GRID CONCEPTS, DISTRIBUTION, COMMUNICATION, METERING, PMU, LOAD FLOW AND ANALYSIS TOOLS

11 topics

cover 100% of all Section A marks asked

- Smart grid basics, active vs passive distribution, enabling technologies
- DG and microgrid, SCADA and HAN/NAN/WAN, smart meter and AMI
- DSM curves, PMU and WAMS
- Load flow, contingency/SSA/SCOPF/DSOPF/congestion multi-agent systems

SECTION B: POWER SYSTEM CONTROL AND STABILITY, POWER QUALITY, INTELLIGENCE, SECURITY, RENEWABLES

10 topics

cover 98% of all Section B marks asked

- Buses and Y_{bus}, frequency control and AGC, reactive power, AVR, SVC
- Rotor-angle stability, swing equation, ω_n proof, EAC + machine numerical
- Voltage stability and VIPI, power quality, ANN/ES/AHP/decision tools
- Interoperability and cyber security, renewables and V2G, barriers and standards

- **Section A repeats strongly:** conventional vs smart grid (A1), active vs passive distribution (A2), the three technology families (A3), DG + microgrid (A4), HAN/NAN/WAN and Bluetooth/ZigBee/WiMAX (A5), AMR vs AMI and meter architecture (A6), DSM curves (A7) and PMU (A8) were each asked in two or three of the three papers.
- **Load flow and analysis tools (A9, A10)** and **multi-agent systems (A11)** close Section A: learn the old-vs-desired load flow table and the definitions of contingency, SSA, SCOPF, DSOPF, congestion.

- **Section B:** AGC and governor (B2), reactive power/AVR/SVC (B3), stability with swing equation and EAC (B4), voltage stability with VIPI (B6), ANN/ES/AHP (B8) and cyber security/false data injection (B9) are the core.
- **Maths is light:** Y_{bus} derivation (B1), the small-disturbance proof $\omega_n = \sqrt{(\pi f_0 P_s/H)}$ (B4) and the transient emf numerical (B5) are each a 6-8 line step table.
- **Always draw:** smart grid architecture, active distribution loop, microgrid, HAN-NAN-WAN, meter-to-utility chain, DSM curves, PMU block diagram, LFC loop, AVR loop, SVC, P-δ curve with equal areas, P-V nose curve, AHP hierarchy.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: SMART GRID CONCEPTS, DISTRIBUTION, COMMUNICATION, METERING, PMU AND ANALYSIS TOOLS: MARKS ASKED PER PAPER

Topic	2023	2022	2020	Years	Marks		Priorit
A1 What a smart grid is: definition, conventional grid vs smart grid, architecture and components, key drivers, stakeholders, need for Bangladesh	10	20	15	3/3	45	Must study	Very like
A3 Enabling technologies: information and communication; sensing, measurement, control and automation; power electronics and energy storage	15	12	15	3/3	42	Must study	Very like
A6 Smart meter, AMR vs AMI, how AMI works, architecture for communication between smart meter and utility server	10	15	15	3/3	40	Must study	Very like
A9 Load flow in smart grids: classical methods, old vs desired	10	10	10	3/3	30	Must study	Very like

load flow, essential features, why needed and challenges							
A4 Distributed generation (DG) and microgrid: definition, advantages, configuration, grid-connected and islanded operation	20	19		2/3	39	Must study	Very like
A7 Demand side management (DSM): load shifting, valley filling, peak clipping (load-shape curves)	15	12		2/3	27	Must study	Very like
A2 Active vs passive distribution network; active distribution management scheme (block diagram)	10	15		2/3	25	Must study	Very like
A5 Smart grid communication: SCADA, HAN/NAN/WAN, and where Bluetooth, ZigBee, WLAN, WPAN and WiMAX are used	15	12	15	3/3	42	Must study	Very likely
A10 Performance analysis tools: contingency analysis, security-constrained OPF, DSOPF, static security assessment (SSA), congestion management; GIS	15	5	20	3/3	40	Must study	Very likely
A8 Phasor measurement unit (PMU) and synchronised phasor measuring system (WAMS)	10	10		2/3	20	Must study	Very likely
A11 Multi-agent systems (MAS): IDAPS and decentralised meter reading	10	10		2/3	20	Must study	Very likely

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: POWER SYSTEM CONTROL AND STABILITY, POWER QUALITY, INTELLIGENCE RENEWABLES: MARKS ASKED PER PAPER

Topic	2023	2022	2020	Years	Marks		Priorit
B8 Computational intelligence and decision tools: ANN, expert systems, AHP, decision support	20	32	20	3/3	72	Must study	Very like

tools, stability tools and computational challenges							
B9 Interoperability and cyber security: threats, false data injection attacks, mitigation	13	7	30	3/3	50	Must study	Very like
B3 Reactive power and voltage control: why the grid needs reactive power, control methods, AVR, SVC, improving P_{max}	28	7	8	3/3	43	Must study	Very like
B11 Barriers and solutions, standards and research areas, the simplest and the most complex thing about the smart grid	13	8	12	3/3	33	Must study	Very like
B10 Renewables and the smart grid: integration challenges, education and training, V2G, sustainable energy options, efficiency and DG control	20	15		2/3	35	Must study	Very like
B7 Power quality: issues in a smart grid, importance of power-quality management and audit	7	8		2/3	15	Must study	Very like
B2 Frequency control: why grid frequency matters, turbine speed governing system, load frequency control loop, AGC	15	21	12	3/3	48	Must study	Very likely
B6 Voltage stability: why instability occurs, preventive control, assessment techniques (P-V, Q-V, VIPI, indices)	12	8	22	3/3	42	Must study	Very likely
B1 Buses and bus types, infinite bus, bus admittance matrix Y_{bus} and proof of $I_{bus} = Y_{bus}V_{bus}$	15	6		2/3	21	Must study	Very likely
B4 Rotor angle stability: steady-state vs transient, swing equation, small-disturbance solution $\omega_n = \sqrt{(\pi f_0 P_s/H)}$, equal area criterion	12	28		2/3	40	Must study	Very like
B5 Numerical: voltage behind transient reactance and transient			10	1/3	10	Good to have	Possi

power-angle equation (with and without saliency)

Section A: Smart grid concepts, distribution, communication, metering, PMU, load flow and analysis tools

questions 1-4 (1-3 in 2020)

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A1 What a smart grid is: definition, conventional grid vs smart grid, architecture and components, key drivers, stakeholders, need for Bangladesh

Must study · Very likely · 100% · Easy

Asked in 3 of 3 papers · 45 marks in total

STUDY THESE

- Smart grid = an electricity network that uses two-way digital communication, sensors and automatic control so that power and information flow both ways between utility and consumer
- Conventional grid: one-way power, centralised generation, few sensors, manual restoration, passive consumers. Smart grid: two-way, distributed generation, self-healing, active consumers
- Drivers: rising demand, ageing infrastructure, renewables, reliability, climate targets, customer choice; stakeholders: generators, transmission and distribution utilities, regulators, consumers/prosumers, vendors, researchers

> 2023 Q1(a) 10 marks
>
> Make a comparison between the conventional power system and a smart grid.

> 2022 Q1(b) 10 marks
>
> Who are the main stakeholders of smart grid systems? Describe their functions.

> 2022 Q1(c) 10 marks

> Mention the key drivers of smart grid. Is a smart grid necessary for Bangladesh? Justify your opinion.

> 2020 Q1(a) 3 marks
> What is a smart grid?

> 2020 Q1(b) 12 marks
> Illustrate the architecture of the smart grid design. What are the main functions of smart grid components?

Full answer: ESE 4007 Study Pack, topic A1.

A3 Enabling technologies: information and communication; sensing, measurement, control and automation; power electronics and energy storage

Must study · Very likely · 100% · Easy

Asked in 3 of 3 papers · 42 marks in total

STUDY THESE

- ICT carries data both ways (wired: fibre, PLC; wireless: ZigBee, Wi-Fi, WiMAX, cellular) with standard protocols and security
- Sensing and measurement: smart meters, PMUs, sensors; control and automation: SCADA, DMS, substation and feeder automation, self-healing
- Power electronics: inverters, FACTS, HVDC, converters for DG; storage: batteries, pumped hydro, flywheels, supercapacitors, SMES - smooth renewables and shift load

> 2023 Q1(c) 15 marks
> Provide an overview of the following technologies required for smart grids: (i) information and communication technologies, (ii) sensing, measurement, control and automation technologies, (iii) power electronics and energy storage.

> 2022 Q2(b) 12 marks
> Describe the roles of sensing, measurement, control and automation technologies for a smart grid.

> 2020 Q1(c) 15 marks
> Provide an overview of the following technological requirements for a smart grid: (i) information and communication technologies, (ii) sensing, measurement,

> control and automation technologies, (iii) power electronics and energy storage systems.

Full answer: ESE 4007 Study Pack, topic A3.

A6 Smart meter, AMR vs AMI, how AMI works, architecture for communication between smart meter and utility server

Must study · Very likely · 100% · Easy

Asked in 3 of 3 papers · 40 marks in total

STUDY THESE

- Smart meter = electronic meter that records energy at short intervals and communicates both ways with the utility
- AMR (automatic meter reading) = one-way, reads the meter automatically for billing. AMI (advanced metering infrastructure) = two-way system: smart meters + communication network + head-end + MDMS, allows remote connect/disconnect, time-of-use pricing, outage alerts
- Path: smart meter (HAN) → data concentrator/collector (NAN) → WAN → head-end system → MDMS → billing, outage, DMS

> 2023 Q3(a) 10 marks
>
> What are the differences between AMR and AMI? How do AMI meters work?

> 2022 Q3(a) 9 marks
>
> What is a smart meter? Using a diagram, show the typical architecture for communication between a smart meter and the utility server.

> 2022 Q3(b) 6 marks
>
> What is advanced metering infrastructure (AMI) in smart grid? Write down the differences between AMR and AMI.

> 2020 Q2(a) 15 marks
>
> Using a diagram show a typical architecture for communication between a smart meter and the utility server.

Full answer: ESE 4007 Study Pack, topic A6.

A9 Load flow in smart grids: classical methods, old vs desired load flow, essential features, why needed and challenges

Must study · Very likely · 100% · Easy

Asked in 3 of 3 papers · 30 marks in total

STUDY THESE

- Load flow finds bus voltages (magnitude and angle) and line flows for given loads and generation; classical methods: Gauss-Seidel, Newton-Raphson, fast decoupled (and DC load flow)
- Old: deterministic, balanced, one snapshot, central generation, transmission only. Desired: handles DG and two-way flow, unbalanced distribution, uncertainty (probabilistic), real-time with PMU data, fast and robust
- Challenges: intermittent renewables, high R/X distribution feeders (convergence), unbalanced three-phase, size, real-time speed

> 2023 Q4(c) 10 marks
>
> Write down the name of the classical methods of load flow solution. What are the essential features of the load flow solution in smart grids?

> 2022 Q4(b) 10 marks
>
> Make a comparison between the old and desired load flow technique in smart grids.

> 2020 Q3(a) 10 marks
>
> Why are load flow studies essential in smart grid? Discuss the challenges of load flow in smart grid.

Full answer: ESE 4007 Study Pack, topic A9.

A4 Distributed generation (DG) and microgrid: definition, advantages, configuration, grid-connected and islanded operation

Must study · Very likely · 83% · Easy

Asked in 2 of 3 papers · 39 marks in total

STUDY THESE

- DG = small generation (a few kW to tens of MW) connected to the distribution network near the load: solar PV, wind, micro-turbines, fuel cells, diesel, CHP
- Advantages: lower losses, better voltage and reliability, deferred network investment, renewables, CHP efficiency, quick to build
- Microgrid = a group of DGs, storage and loads with its own controller that can run connected to the grid or alone (islanded) via a static switch at the point of common coupling

> 2023 Q2(a) 10 marks
>
> Define the term "distributed generation (DG)". What are the advantages of DG?

> 2023 Q2(b) 10 marks
>
> Draw the typical configuration of a micro-grid. Describe the grid-connected and islanded operation of the micro-grid.

> 2022 Q2(a) 11 marks
>
> Define the term "distributed generation (DG)". What are the advantages of DG?

> 2022 Q3(d) 8 marks
>
> Draw the configuration of a typical microgrid and describe its operation.

Full answer: ESE 4007 Study Pack, topic A4.

A7 Demand side management (DSM): load shifting, valley filling, peak clipping (load-shape curves)

Must study · Very likely · 83% · Easy

Asked in 2 of 3 papers · 27 marks in total

STUDY THESE

- DSM = utility actions that change the amount or timing of customers' electricity use to flatten the load curve
- Peak clipping: cut the peak (direct load control). Valley filling: add load at night/off-peak (EV charging, storage). Load shifting: move load from peak to off-peak (time-of-use tariff, thermal storage)
- Also: strategic conservation, strategic load growth, flexible load shape

> 2023 Q3(b) 15 marks
>
> What is meant by demand side management? Describe how the following techniques modify the load consumption patterns: (i) load shifting, (ii) valley filling, (iii) peak clipping.

> 2022 Q3(c) 12 marks
>
> What is meant by demand side management? Describe how the following techniques modify the load consumption patterns: (i) load shifting, (ii) valley filling, (iii) peak clipping.

Full answer: ESE 4007 Study Pack, topic A7.

A2 Active vs passive distribution network; active distribution management scheme (block diagram)

Must study · Very likely · 83% · Easy

Asked in 2 of 3 papers · 25 marks in total

STUDY THESE

- Passive network: power flows one way from substation to loads, no generation, simple protection, little monitoring ("fit and forget")
- Active network: DG, storage and controllable loads connected; power can flow both ways; it is monitored and controlled in real time
- Active distribution management: measurements → state estimation → control functions (voltage control, DG dispatch, reconfiguration, demand response) → commands to devices

> 2023 Q1(b) 10 marks
>
> What are the differences between an active and a passive distribution network? Draw the block diagram of an active distribution system and describe its operation.

> 2022 Q1(a) 15 marks
>
> What are the differences between an active and a passive distribution network? Describe the operation of an active distribution management scheme with necessary block diagrams.

Full answer: ESE 4007 Study Pack, topic A2.

A5 Smart grid communication: SCADA, HAN/NAN/WAN, and where Bluetooth,

Must study · Very likely · 100% · Medium

ZigBee, WLAN, WPAN and WiMAX are used

Asked in 3 of 3 papers · 42 marks in total

STUDY THESE

- SCADA = supervisory control and data acquisition: RTUs/IEDs in the field, a communication network and a master station with HMI, to monitor and control the grid remotely
- Three network levels: HAN (home: meter, appliances, ~10-100 m), NAN (neighbourhood: many meters to a data collector, ~1-10 km), WAN (wide area: backbone to the utility control centre, 10-100+ km)
- Bluetooth/WPAN and ZigBee inside the home; WLAN (Wi-Fi) in homes, substations and NAN mesh; WiMAX for NAN/WAN backhaul and distribution automation

> `2023` `Q2(c)` `15 marks`
>
> What is meant by SCADA? What are the advantages of a SCADA integrated system? Show the possible communication infrastructure for the smart grid including HAN, NAN and WAN.

> `2022` `Q2(c)` `12 marks`
>
> Define HAN, NAN and WAN. Compare the conventional and smart meter. Write down the application areas of the following communication technologies in smart grid: Bluetooth, WiMAX and WLAN.

> `2020` `Q2(b)` `15 marks`
>
> Write down the application areas of the following communication technologies in smart grid: Bluetooth, ZigBee, WLAN, WPAN, WiMAX.

Full answer: ESE 4007 Study Pack, topic A5.

`A10`

Performance analysis tools: contingency analysis, security-constrained OPF, DSOPF, static security assessment (SSA), congestion management; GIS

Must study · Very likely · 100% · Medium

Asked in 3 of 3 papers · 40 marks in total

STUDY THESE

- Contingency analysis = "what if" a line/generator trips (N-1): check overloads and voltage limits; SSA uses it to classify the system as secure or insecure
- OPF = minimise cost subject to power flow and limits; SCOPF adds the constraint that the system stays within limits after each contingency; DSOPF = dynamic stochastic OPF (handles time and uncertainty, adaptive)
- Congestion = a line reaching its limit, managed by re-dispatch, FACTS, pricing; GIS and Google maps show assets and outages on maps for planning and crews

> 2023 Q4(a) 10 marks
>
> Why is contingency analysis important for power system studies? Describe the term "security-constrained optimal power flow" with an example.

> 2023 Q4(b) 5 marks
>
> Write down the necessity of GIS and Google mapping tools in smart grids.

> 2022 Q4(c) 5 marks
>
> Write a short note on static security assessment (SSA) of smart power systems.

> 2020 Q3(b) 20 marks
>
> Provide an overview of the following performance analysis tools for smart grid design: (i) DSOPF, (ii) SSA, (iii) congestion, (iv) contingency.

Full answer: ESE 4007 Study Pack, topic A10.

A8 Phasor measurement unit (PMU) and synchronised phasor measuring system (WAMS)

Must study · Very likely · 83% · Medium

Asked in 2 of 3 papers · 20 marks in total

STUDY THESE

- PMU measures voltage and current phasors (magnitude and angle) and frequency, time-stamped by GPS to 1 µs, 30-60 times per second
- Inside: anti-aliasing filter → A/D converter (sampling locked to GPS via a phase-locked oscillator) → microprocessor (DFT phasor computation) → communication
- WAMS: many PMUs → PDC (phasor data concentrator) → super PDC at the control centre → applications (state estimation, stability monitoring, protection)

> 2023 Q3(c) 10 marks
>
> What are the key features of a phasor measurement unit (PMU)? Explain the operation of a PMU with necessary block diagrams.

> 2022 Q4(a) 10 marks
>
> What is a phasor measurement unit in smart grid? Discuss a synchronized phasor measuring system with conceptual diagram.

Full answer: ESE 4007 Study Pack, topic A8.

A11 Multi-agent systems (MAS): IDAPS and decentralised meter reading

Must study · Very likely · 83% · Medium

Asked in 2 of 3 papers · 20 marks in total

STUDY THESE

- Agent = software that senses its environment, decides and acts on its own; MAS = many agents that cooperate to reach a goal without one central controller
- IDAPS (intelligent distributed autonomous power system): control agent, DER agent, user (load) agent and database agent; islands the microgrid on a fault and serves critical loads
- Decentralised meter reading: meter agents → collector agents → billing/database agent, talking by messages; no single point of failure

> 2023 Q4(d) 10 marks
>
> Design a decentralized meter reading system using the multi-agent approach.

> 2022 Q4(d) 10 marks
>
> What is meant by a multi-agent system (MAS)? Develop an intelligent distributed autonomous power system (IDAPS) using the MAS approach.

Full answer: ESE 4007 Study Pack, topic A11.

Section B: Power system control and stability, power quality, intelligence, security, renewables

questions 5-8 (4-6 in 2020)

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B8 Computational intelligence and decision tools: ANN, expert systems, AHP, decision support tools, stability tools and computational challenges

Must study · Very likely · 100% · Easy

Asked in 3 of 3 papers · 72 marks in total

STUDY THESE

- ANN (artificial neural network) learns from data: load and renewable forecasting, fault detection, security assessment, state estimation
- Expert system = knowledge base of rules + inference engine; used for fault diagnosis, alarm processing, restoration, operator advice. Decision support tools combine data, models and optimisation for planning and operation
- AHP (analytical hierarchy process) ranks options: goal → criteria → alternatives, pairwise comparisons, weights, consistency check

> 2023 Q7(b) 10 marks
>
> What is meant by ANN? How does the ANN facilitate the assessment of smart grid performance?

> 2023 Q8(b) 10 marks
>
> How do expert systems (ES) enhance decision-making and operational efficiency in smart grid? Discuss.

> 2022 Q7(b) 10 marks
>
> How can decision support tools be used in smart grid planning and operations? Describe.

> 2022 Q7(c) 10 marks
>
> What is the role of the Analytical Hierarchical Programming (AHP) method in smart grid management? Explain with necessary block diagram.

> 2022 Q8(d) 12 marks
>
> Describe the computational challenges faced by researchers and engineers for the design, planning and operations of smart power system.

> 2020 Q5(a) 10 marks
>
> What are the stability analysis tools for smart grid? Write the reasons of computational challenges in smart grids.

> 2020 Q5(c) 10 marks

How does an Artificial Neural Network (ANN) improve smart grid technology? Illustrate the Analytical Hierarchical Programming (AHP) method.

Full answer: ESE 4007 Study Pack, topic B8.

B9 Interoperability and cyber security: threats, false data injection attacks, mitigation

Must study · Very likely · 100% · Easy

Asked in 3 of 3 papers · 50 marks in total

STUDY THESE

- Interoperability = devices and systems from different vendors can exchange and use information (common standards: IEC 61850, CIM, DNP3, IEEE 2030)
- Cyber threats: more connected devices = more doors for attackers (malware, DoS, man-in-the-middle, meter tampering); goals: confidentiality, integrity, availability
- False data injection: attacker changes meter/sensor data so that state estimation accepts wrong values without triggering bad-data detection → wrong dispatch, overloads, blackouts. Mitigate: protect key meters, PMUs, encryption, authentication, anomaly detection

`2023 Q7(c) 13 marks`

What is interoperability in smart grid environment? How can false data injection impact the stability and reliability of smart grid operations, and what measures can be taken to mitigate these risks effectively?

`2022 Q7(d) 7 marks`

Explain why cyber security is of prime importance in smart grids and how the risk of cyber-attack can be mitigated.

`2020 Q6(a) 12 marks`

What is interoperability in smart grid environment? How is cyber security a threat to smart grids?

`2020 Q6(b) 12 marks`

What is false data injection attack in the smart grid? (AGC part in B2)

`2020 Q6(c) 6 marks`

What are the main mitigation approaches to cyber security risk?

Full answer: ESE 4007 Study Pack, topic B9.

B3 Reactive power and voltage control: why the grid needs reactive power, control methods, AVR, SVC, improving P_{max}

Must study · Very likely · 100% · Easy

Asked in 3 of 3 papers · 43 marks in total

STUDY THESE

- Reactive power (Q) builds the magnetic fields of motors and transformers and holds up voltage; voltage depends mainly on Q, frequency on P
- Q control devices: generator excitation (AVR), shunt capacitors and reactors, synchronous condensers, tap-changing transformers, SVC, STATCOM, series capacitors
- P_{max} = EV/X: raise it by higher E (excitation), higher V (voltage support), lower X (parallel lines, series capacitors, bundled conductors, FACTS)

> 2023 Q5(b) 13 marks
>
> How can the maximum power output (P_{max}) of a generator be effectively improved? What are the primary functions of a thyristor-based SVC (static var compensator) in power system? Explain with diagram.

> 2023 Q6(c) 15 marks
>
> What are the primary methods and devices used to control reactive power in power systems? How does the AVR (automatic voltage regulator) maintain voltage stability in smart grid? (loop part in B2)

> 2022 Q6(d) 7 marks
>
> Why does the grid need reactive power? Discuss.

> 2020 Q4(c) 8 marks
>
> What is the necessity of reactive power control in grid? Write the different methods of reactive power control.

Full answer: ESE 4007 Study Pack, topic B3.

B11 Barriers and solutions, standards and research areas, the simplest and the most complex thing about the smart grid

Must study · Very likely · 100% · Easy

Asked in 3 of 3 papers · 33 marks in total

STUDY THESE

- Barriers: high cost and unclear payback, regulation and tariffs, legacy systems, lack of standards/interoperability, cyber security and privacy, skills shortage, consumer acceptance
- Standards (IEEE 2030, IEC 61850, IEC 61968/61970 CIM, NIST framework) let equipment work together, protect investment, ensure safety and security
- Simplest: the idea - add sensing, communication and control to the grid. Complex: making millions of devices, markets and people work together securely in real time

> 2023 Q8(a) 13 marks
>
> What are the primary barriers to the development of smart grids, and what solutions can be implemented to overcome these obstacles? Discuss.

> 2022 Q8(c) 8 marks
>
> Why are smart grid standards necessary? What would be the suitable areas of research in smart grid?

> 2020 Q4(a) 12 marks
>
> What are the simplest thing and the complex thing about the smart grid? (voltage stability part in B6)

Full answer: ESE 4007 Study Pack, topic B11.

B10 Renewables and the smart grid: integration challenges, education and training, V2G, sustainable energy options, efficiency and DG control

Must study · Very likely · 83% · Easy

Asked in 2 of 3 papers · 35 marks in total

STUDY THESE

- Renewables are variable and uncertain, inverter-based (low inertia), distributed; research challenges: forecasting, storage, stability, protection, market design
- V2G (vehicle-to-grid): parked electric vehicles charge when power is cheap and send power back at peaks or for frequency regulation

- Smart grid raises efficiency by cutting losses (Volt/VAR optimisation), peak reduction (DSM), better asset use, and controlled DG close to loads

> 2023 Q7(a) 12 marks
>
> What are some current research challenges in integrating renewable energy sources into the smart grid infrastructure? How does education and training contribute to the successful implementation of smart grid?

> 2023 Q6(b) 8 marks
>
> Write a short note on vehicle-to-grid (V2G) technology.

> 2022 Q8(a) 8 marks
>
> How does a smart grid improve efficiency? What are the benefits of distributed generation control in smart grid?

> 2022 Q8(b) 7 marks
>
> Give an overview of sustainable energy options for the smart grid.

Full answer: ESE 4007 Study Pack, topic B10.

B7 Power quality: issues in a smart grid, importance of power-quality management and audit

Must study · Very likely · 83% · Easy

Asked in 2 of 3 papers · 15 marks in total

STUDY THESE

- Power quality = how close voltage and current are to a perfect 50 Hz sine wave at rated value
- Issues: sags (dips), swells, interruptions, transients/surges, harmonics (from inverters, drives, LEDs), flicker, unbalance, frequency deviation
- PQ audit = measure and analyse PQ at a site to find problems, causes and fixes (filters, UPS, DVR, STATCOM); protects sensitive equipment and saves money

> 2023 Q5(c) 7 marks
>
> Why is power quality management in smart grid so important?

> 2022 Q5(b) 8 marks
>
> What are the power quality issues with smart grid? What is the importance of power quality audit in smart grid? Discuss.

Full answer: ESE 4007 Study Pack, topic B7.

B2 Frequency control: why grid frequency matters, turbine speed governing system, load frequency control loop, AGC

Must study · Very likely · 100% · Medium

Asked in 3 of 3 papers · 48 marks in total

STUDY THESE

- Frequency stays at 50 Hz only when generation = load; a load increase slows the machines and frequency falls
- Speed governor (fly-ball/electronic) senses speed and opens the steam/water valve: primary control with droop R (frequency falls a little, output rises)
- AGC (automatic generation control) = secondary control: integral of the area control error (ACE) resets frequency to 50 Hz and tie-line flows to schedule; essential with variable renewables

> `2023` `Q6(c)` `15 marks`
>
> Draw and discuss the simple feedback loop block diagram of generator and load. What are the primary methods and devices used to control reactive power in power systems? And how does the AVR maintain the voltage stability in smart grid? (reactive power part in B3)

> `2022` `Q5(c)` `12 marks`
>
> Why is power grid frequency important? How does the turbine speed governing system work? Explain with schematic diagram.

> `2022` `Q5(d)` `9 marks`
>
> What is Automatic Generation Control (AGC) in power system? "AGC is essential in smart grid" - justify.

> `2020` `Q6(b)` `12 marks`
>
> What is false data injection attack in the smart grid? "Automatic Generation Control (AGC) is essential for power system" - justify. (attack part in B9)

Full answer: ESE 4007 Study Pack, topic B2.

B6 Voltage stability: why instability occurs, preventive control,

Must study · Very likely · 100% · Medium

assessment techniques (P-V, Q-V, VIPI, indices)

Asked in 3 of 3 papers · 42 marks in total

STUDY THESE

- Voltage instability = voltages fall uncontrollably because the network cannot supply the reactive power the loads demand (heavy load, long lines, Q limits of generators, tap-changer action, motor loads)
- Preventive control: reactive reserves, capacitor/SVC switching, generator voltage set-points, tap-changer blocking, load shedding, re-dispatch
- Assessment: P-V (nose) curve, Q-V curve, modal/eigenvalue analysis, continuation power flow, voltage stability indices (L-index), VIPI (voltage instability proximity index based on the angle between two load-flow solution vectors)

> 2023 Q8(c) 12 marks
>
> Mention the names of different voltage stability assessment techniques. Describe one of them.

> 2022 Q7(a) 8 marks
>
> Why does voltage instability occur in power grid? Discuss the voltage instability preventive control scheme for the power grid.

> 2020 Q4(a) 12 marks
>
> What are the simplest thing and the complex thing about the smart grid? Write about the major reasons for voltage stability problems in power system. (first part in B11)

> 2020 Q5(b) 10 marks
>
> Classify the techniques of voltage stability assessment. Describe the VIPI method for voltage stability assessment.

Full answer: ESE 4007 Study Pack, topic B6.

B1 Buses and bus types, infinite bus, bus admittance matrix Y_{bus} and proof of $I_{bus} = Y_{bus}V_{bus}$

Must study · Very likely · 83% · Medium

Asked in 2 of 3 papers · 21 marks in total

STUDY THESE

- Bus = a node where lines, generators and loads meet. Three types for load flow: slack (V, δ given), PV/generator (P, |V| given), PQ/load (P, Q given)
- Infinite bus = a bus whose voltage and frequency stay constant whatever power you inject (huge grid, zero internal impedance, infinite inertia)
- Y_{bus}: diagonal Y_{ii} = sum of all admittances connected to bus i; off-diagonal Y_{ij} = − (admittance between i and j); from KCL at each bus you get I = YV

> 2023 Q5(a) 15 marks
>
> What are the bus and infinite bus in power system? Describe a method to derive the bus admittance matrix in smart power system and prove that $I_{bus} = Y_{bus}V_{bus}$.

> 2022 Q5(a) 6 marks
>
> How many buses are there in the power system? What is the significance of different types of buses in the power system?

Full answer: ESE 4007 Study Pack, topic B1.

B4 Rotor angle stability: steady-state vs transient, swing equation, small-disturbance solution $\omega_n = \sqrt{(\pi f_0 P_s/H)}$, equal area criterion

Must study · Very likely · 83% · Hard

Asked in 2 of 3 papers · 40 marks in total

STUDY THESE

- Steady-state stability: survive small, slow changes; transient stability: survive a large sudden disturbance (fault, line trip)
- Swing equation: $(H/\pi f_0)\, d^2\delta/dt^2 = P_m - P_e$; the rotor angle accelerates when mechanical input exceeds electrical output
- Small disturbance: linearise, $P_s = P_{max}\cos\delta_0$; oscillation at $\omega_n = \sqrt{(\pi f_0 P_s/H)}$. Equal area criterion: stable if decelerating area A_2 can equal accelerating area A_1

> 2023 Q6(a) 12 marks
>
> In power system transient stability analysis, what principle does the equal area criterion (EAC) assess? Describe with necessary schematic diagram.

> 2022 Q6(a) 8 marks
>
> Distinguish between transient and steady-state stability and discuss the need for performing stability analyses of power system.

> 2022 Q6(b) 6 marks
>
> Elucidate the importance and role of the swing equation in power system stability analysis.

> 2022 Q6(c) 14 marks
>
> Discuss the possible solutions of the disturbed motion of a synchronous generator connected to an infinite bus, when subjected to a small disturbance represented by $\Delta\delta$, and prove that $\omega_n = \sqrt{(\pi f_0 P_s/H)}$.

Full answer: ESE 4007 Study Pack, topic B4.

B5 Numerical: voltage behind transient reactance and transient power-angle equation (with and without saliency)

Good to have · Possible · 17% · Hard

Asked in 1 of 3 papers · 10 marks in total

STUDY THESE

- $I = P/(V \cdot pf)$ at angle $-\cos^{-1}pf$; $E' = V + jx'_dI$ (no saliency), $P = (E'V/x'_d)\sin\delta$
- With saliency: $E_Q = V + jx_qI$ gives δ; $I_d = |I|\sin(\delta + \phi)$; $E'_q = |E_Q| - (x_q - x'_d)I_d$
- $P = (E'_qV/x'_d)\sin\delta + (V^2/2)(1/x_q - 1/x'_d)\sin2\delta$

> 2020 Q4(b) 10 marks
>
> A synchronous machine has $x_d = 1.0$, $x_q = 0.6$, $x'_d = 0.3$ pu and negligible armature resistance. It is connected directly to an infinite bus of 1.0 pu and delivers 0.5 pu real power at 0.8 pf lagging. Determine the voltage behind transient reactance and the transient power-angle equation (i) neglecting saliency, (ii) including saliency.

Full answer: ESE 4007 Study Pack, topic B5.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
A/D	analog-to-digital

ACE	area control error
ACL	agent communication language
ADB	Asian Development Bank
ADMS	advanced distribution management system
AGC	automatic generation control
AHP	analytical hierarchy process (the paper calls it analytical hierarchical programming)
AI	artificial intelligence
AMI	advanced metering infrastructure
AMR	automatic meter reading
ANN	artificial neural network
ANSI	American National Standards Institute
AVR	automatic voltage regulator
BAN	building area network
BERC	Bangladesh Energy Regulatory Commission
BREB	Bangladesh Rural Electrification Board
CAES	compressed air energy storage
CHP	combined heat and power
CI	consistency index
CIA	confidentiality, integrity, availability
CIM	common information model
CIP	critical infrastructure protection
CIS	customer information system
CO_2	carbon dioxide
COSEM	Companion Specification for Energy Metering
CPF	continuation power flow
CR	consistency ratio
CT	current transformer
DER	distributed energy resource
DESCO	Dhaka Electric Supply Company

DFT	discrete Fourier transform
DG	distributed generation
DLMS	Device Language Message Specification
DLR	dynamic line rating
DMS	distribution management system
DMZ	demilitarised zone (buffer network)
DNP3	Distributed Network Protocol 3
DoS	denial of service
DPDC	Dhaka Power Distribution Company
DR	demand response
DSL	digital subscriber line
DSM	demand side management
DSOPF	dynamic stochastic optimal power flow
DST	decision support tool
DVR	dynamic voltage restorer
EAC	equal area criterion
EMS	energy management system
ES	expert system
EV	electric vehicle
FACTS	flexible AC transmission systems
FAN	field area network
FDI	false data injection
FDLF	fast decoupled load flow
FIPA	Foundation for Intelligent Physical Agents
FLISR	fault location, isolation and service restoration
G2V	grid-to-vehicle
GIS	geographic information system
GPS	Global Positioning System
GS	Gauss-Seidel

HAN	home area network
HES	head-end system
HMI	human-machine interface
HVDC	high-voltage direct current
Hz	hertz (cycles per second)
ICT	information and communication technology
IDAPS	intelligent distributed autonomous power system
IEC	International Electrotechnical Commission
IED	intelligent electronic device
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
IPP	independent power producer
IT	information technology
JADE	Java Agent DEvelopment framework
KCL	Kirchhoff's current law
KVL	Kirchhoff's voltage law
kW	kilowatt
kWh	kilowatt-hour
LED	light-emitting diode
LFC	load frequency control
LNG	liquefied natural gas
LTE	Long Term Evolution (4G cellular)
MAS	multi-agent system
MDMS	meter data management system
MGCC	microgrid central controller
MTU	master terminal unit (master station)
MVA	mega volt-ampere
MW	megawatt
NAN	neighbourhood area network

NERC	North American Electric Reliability Corporation
NESCO	Northern Electricity Supply Company
NIST	National Institute of Standards and Technology (USA)
NLDC	National Load Despatch Centre
NR	Newton-Raphson
OLTC	on-load tap changer
OMS	outage management system
OpenADR	Open Automated Demand Response
OPF	optimal power flow
OT	operational technology
PCC	point of common coupling
PDC	phasor data concentrator
pf	power factor
PGCB	Power Grid Company of Bangladesh (now Power Grid Bangladesh PLC)
PLC	programmable logic controller (or power line communication, by context)
PMU	phasor measurement unit
PPS	pulse per second
PQ	power quality (also PQ bus = P and Q specified)
PSS	power system stabiliser
PT	potential (voltage) transformer
pu	per unit (value / base value)
PV	photovoltaic (solar cells); also PV bus = P and |V| specified
RF	radio frequency
RI	random index
ROCOF	rate of change of frequency
RTU	remote terminal unit
SCADA	supervisory control and data acquisition
SCOPF	security-constrained optimal power flow
SG	smart grid

SMES	superconducting magnetic energy storage
SMPS	switched-mode power supply
SREDA	Sustainable and Renewable Energy Development Authority
SSA	static security assessment
STATCOM	static synchronous compensator
SVC	static var compensator
TCR	thyristor-controlled reactor
TCSC	thyristor-controlled series capacitor
THD	total harmonic distortion
Tk	Taka
TLS	Transport Layer Security
ToU	time of use
TSC	thyristor-switched capacitor
UPFC	unified power flow controller
UPS	uninterruptible power supply
UTC	Coordinated Universal Time
V2G	vehicle-to-grid
V2H	vehicle-to-home
VEE	validation, editing and estimation
VFD	variable frequency drive
VIPI	voltage instability proximity index
VPN	virtual private network
WAMPAC	wide area monitoring, protection and control
WAMS	wide area monitoring system
WAN	wide area network
WiMAX	Worldwide Interoperability for Microwave Access (IEEE 802.16)
WLAN	wireless local area network (Wi-Fi)
WPAN	wireless personal area network
WZPDCL	West Zone Power Distribution Company Ltd

Built from the scanned ESE 4007 papers (2020, 2022, 2023). Question text is cleaned up from the scans.